Name: Deras Pangestu Gusti (08)

Class: SIB 1G

NIM: 254107060114

JOBSHEET
Case Study 1

1. Objective

- Students are able to solve case study problems by applying the concepts of class, object, and array of objects.
- Students are able to apply sorting and searching algorithms according to case studies.

2. Practicum

Processing Time: 120 minutes

Work Instructions:
Each student **will code a program** based on a given case study. Ensure the program runs smoothly, implements all requested features, and is dynamic (using menus to select features).

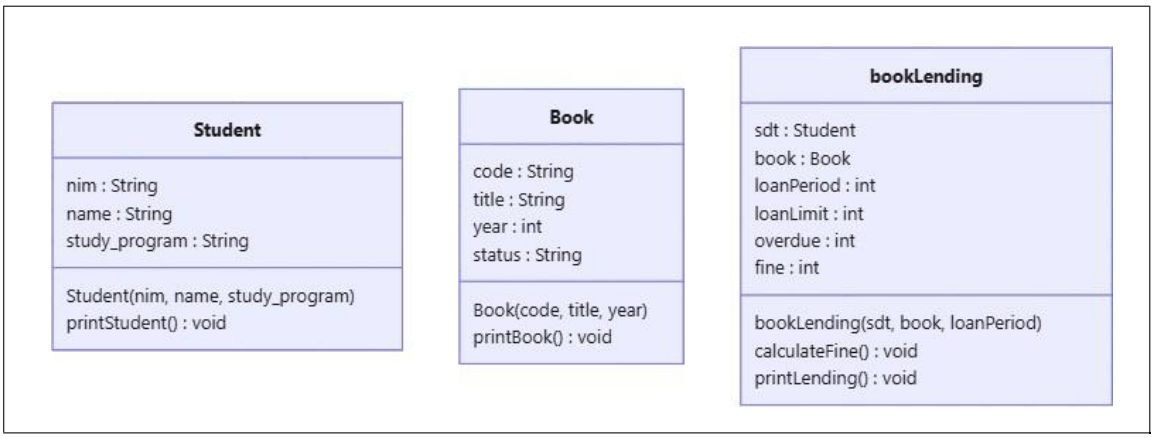
Case Study Questions:
Case Study: JTI Reading Room Book Loan Management System
Students frequently borrow books from the JTI reading room for assignments or theses. The reading room administrator needs a simple system to record book, student, and loan transaction data. This system must also be able to automatically calculate late fees based on the length of the loan. The system must have the following features:

a. Displays book and student data.
b. Displays book borrowing data.
c. Calculate the number of days late and the total fine.
d. Sort loan data by the largest fine amount.
e. Search for loan data based on student NIM.

Provision:

- The maximum limit for borrowing books is 5 days.
- The late fee is Rp. 2,000 per day
- If not late, then the fine = 0

Create a Java program using the following class diagram:



Use the class diagram above to create a book lending management system for the JTI reading room. You can add new classes, attributes, or methods to complete the case study.

The following data can be used as initial data in the program. All data is stored in the form of an array of objects and does not require user input.

a. Student Data

NIM	Name	Study Program
22001	Andi	Informatics Engineering
22002	Budi	Informatics Engineering
22003	Image	Business Information Systems

b. Book Data

Code	Title	Year
B001	Algorithm	2020
B002	Database	2019
B003	Programming	2021
B004	Physics	2024

c. Loan Data

NIM	Name	Book title	Loan Period (days)
22001	Andi	Algorithm	7
22002	Budi	Database	3
22003	Image	Programming	10
22003	Image	Physics	6
22001	Andi	Database	4

The following is an example of the results of running the program:

Menu 1:

```
=== JTI LIBRARY BORROWING SYSTEM ===
1. Display Students
2. Display Books
3. Display Borrowings
4. Sort by Fines
5. Search by Student ID
0. Exit
Select: 1

Student List:
NIM: 22001 | Name: Andi | Study program: Teknik Informatika
NIM: 22002 | Name: Budi | Study program: Teknik Informatika
NIM: 22003 | Name: Citra | Study program: Sistem Informasi Bisnis
```

Menu 2:

```
=== JTI LIBRARY BORROWING SYSTEM ===
1. Display Students
2. Display Books
3. Display Borrowings
4. Sort by Fines
5. Search by Student ID
0. Exit
Select: 2

Book List:
Code: B001 | Title: Algoritma | Year: 2020
Code: B002 | Title: Basis Data | Year: 2019
Code: B003 | Title: Pemrograman | Year: 2021
Code: B004 | Title: Fisika | Year: 2024
```

Menu 3:

```
=== JTI LIBRARY BORROWING SYSTEM ===
1. Display Students
2. Display Books
3. Display Borrowings
4. Sort by Fines
5. Search by Student ID
0. Exit
Select: 3

Loan Data:
Andi | Algoritma | Loan Period (days): 7 | Late return of books: 2 | fine: 4000
Budi | Basis Data | Loan Period (days): 3 | Late return of books: 0 | fine: 0
Citra | Pemrograman | Loan Period (days): 10 | Late return of books: 5 | fine: 10000
Citra | Fisika | Loan Period (days): 6 | Late return of books: 1 | fine: 2000
Andi | Basis Data | Loan Period (days): 4 | Late return of books: 0 | fine: 0
```

Menu 4:

```
=== JTI LIBRARY BORROWING SYSTEM ===
1. Display Students
2. Display Books
3. Display Borrowings
4. Sort by Fines
5. Search by Student ID
0. Exit
Select: 4

After sorting (Largest fine):
Citra | Pemrograman | Loan Period (days): 10 | Late return of books: 5 | fine: 10000
Andi | Algoritma | Loan Period (days): 7 | Late return of books: 2 | fine: 4000
Citra | Fisika | Loan Period (days): 6 | Late return of books: 1 | fine: 2000
Budi | Basis Data | Loan Period (days): 3 | Late return of books: 0 | fine: 0
Andi | Basis Data | Loan Period (days): 4 | Late return of books: 0 | fine: 0
```

Menu 5:

```
=== JTI LIBRARY BORROWING SYSTEM ===
1. Display Students
2. Display Books
3. Display Borrowings
4. Sort by Fines
5. Search by Student ID
0. Exit
Select: 5
Enter NIM: 22003
Citra | Pemrograman | Loan Period (days): 10 | Late return of books: 5 | fine: 10000
Citra | Fisika | Loan Period (days): 6 | Late return of books: 1 | fine: 2000
```

Assessment Rubric

Assessment Aspects	Max Score	Score	Criteria
1. Class and Object Implementation (25%)	25	20-25	All classes (Student, Book, Loan) are complete (attributes, constructors, methods) and object relationships are correct.
		10-19	The class already exists but there are still deficiencies in the attributes/methods or the object relationships are not quite right.
		0-9	The class does not conform to the specification or does not use the object concept correctly.
2. Array of Object Implementation (15%)	15	12-15	Arrays of objects are used correctly and the relationships between objects are correct.
		6-11	Using an array of objects but still has implementation errors.
		0-5	Not using array of objects or incorrect implementation.
3. Implementation of Calculation Method (20%)	20	16-20	Calculation methods such as hitungDenda() are correct, the logic is correct, and the results are stored in attributes.
		8-15	The calculations exist but there are logical errors or the results are inaccurate.

		0-7	There is no calculation method or it is not running.
4. Sorting Implementation (20%)			
a. Types of Algorithms	5	5	Using Insertion Sort
		4	Using Selection Sort
		3	Using Bubble Sort
		0	No sorting
b. Sorting Truth	15	12-15	Data is sorted correctly according to criteria.
		6-11	Partially sorted / there are minor errors.
		0-5	Out of order or wrong.
5. Searching Implementation (20%)			
a. Types of Algorithms	5	5	Using Binary Search
		3	Using Sequential Search
		0	No searching
b. Searching Truth	15	12-15	Search is always correct for all conditions (found & not found).
		6-11	The search is running but there are errors on some conditions.
		0-5	Search does not work or results are incorrect.
Total	**100**		

```java
package com.cm1;

public class Student
{
    String nim, name, study_program;

    Student (String nim, String name, String study_program)
    {
        this.nim = nim;
        this.name = name;
        this.study_program = study_program;
    }

    void printStudent()
    {
        System.out.println("NIM: " + nim + " | Name: " + name + " | Study program: " + study_program);
    }
}
```

```java
package com.cm1;

public class Book
{
    String code, title, status;
    int year;

    Book(String code, String title, int year)
    {
        this.code = code;
        this.title = title;
        this.year = year;
        this.status = "Available";
    }

    void printBook()
    {
        System.out.println("Code: " + code + " | Title: " + title + " | Year: " + year);
    }
}
```

```java
package com.cm1;

public class BookLending
{
    Student sdt;
    Book book;
    int loanPeriod, loanLimit, overdue, fine;

    BookLending(Student sdt, Book book, int loanPeriod)
    {
        this.sdt = sdt;
        this.book = book;
        this.loanPeriod = loanPeriod;
        this.loanLimit = 5;
        calculateFine();
    }

    void calculateFine()
    {
        overdue = (loanPeriod > loanLimit) ? loanPeriod - loanLimit : 0;
        fine = overdue * 2000;
    }

    void printLending()
    {
        System.out.println(sdt.name + " | " + book.title +
            " | Loan Period (days): " + loanPeriod +
            " | Late return of books: " + overdue +
            " | fine: " + fine);
    }
}
```

```java
package com.cm1;

import java.util.Scanner;

public class Main
{

    static int nimCompare(String a, String b)
    {
        int lenA = a.length(), lenB = b.length();
        int minLen = (lenA < lenB) ? lenA : lenB;
        for (int i = 0; i < minLen; i++)
        {
            if (a.charAt(i) != b.charAt(i))
                return a.charAt(i) - b.charAt(i);
        }
        return lenA - lenB;
    }

    static boolean nimEquals(String a, String b)
    {
        if (a.length() != b.length()) return false;
        for (int i = 0; i < a.length(); i++)
        {
            if (a.charAt(i) != b.charAt(i)) return false;
        }
        return true;
    }

    static void insertionSort(BookLending[] arr)
    {
        for (int i = 1; i < arr.length; i++)
        {
            BookLending key = arr[i];
            int j = i - 1;
            while (j >= 0 && arr[j].fine < key.fine)
            {
                arr[j + 1] = arr[j];
                j--;
            }
            arr[j + 1] = key;
        }
    }

```

```java
    static void binarySearchByNIM(BookLending[] loans, String targetNIM)
    {
        // Copy and sort by NIM for binary search
        BookLending[] sorted = loans.clone();
        // Insertion sort by NIM string
        for (int i = 1; i < sorted.length; i++)
        {
            BookLending key = sorted[i];
            int j = i - 1;
            while (j >= 0 && nimCompare(sorted[j].sdt.nim, key.sdt.nim) > 0)
            {
                sorted[j + 1] = sorted[j];
                j--;
            }
            sorted[j + 1] = key;
        }

        int lo = 0, hi = sorted.length - 1;
        int firstFound = -1;
        while (lo <= hi)
        {
            int mid = (lo + hi) / 2;
            int cmp = nimCompare(sorted[mid].sdt.nim, targetNIM);
            if (cmp == 0) { firstFound = mid; hi = mid - 1; }
            else if (cmp < 0) lo = mid + 1;
            else hi = mid - 1;
        }

        if (firstFound == -1)
        {
            System.out.println("NIM " + targetNIM + " not found.");
            return;
        }

        for (int i = firstFound; i < sorted.length && nimEquals(sorted[i].sdt.nim, targetNIM); i++)
        {
            sorted[i].printLending();
        }
    }

```

```java
    public static void main(String[] args) {
        // --- Data ---
        Student[] students =
        {
            new Student("22001", "Andi",  "Informatics Engineering"),
            new Student("22002", "Budi",  "Informatics Engineering"),
            new Student("22003", "Citra", "Business Information Systems")
        };

        Book[] books =
        {
            new Book("B001", "Algorithm",    2020),
            new Book("B002", "Database",     2019),
            new Book("B003", "Programming",  2021),
            new Book("B004", "Physics",      2024)
        };

        BookLending[] loans =
        {
            new BookLending(students[0], books[0], 7),
            new BookLending(students[1], books[1], 3),
            new BookLending(students[2], books[2], 10),
            new BookLending(students[2], books[3], 6),
            new BookLending(students[0], books[1], 4)
        };

        Scanner sc = new Scanner(System.in);
        int choice;

```

```java
        do
        {
            System.out.println("\n=== JTI LIBRARY BORROWING SYSTEM ===");
            System.out.println("1. Display Students");
            System.out.println("2. Display Books");
            System.out.println("3. Display Borrowings");
            System.out.println("4. Sort by Fines");
            System.out.println("5. Search by Student ID");
            System.out.println("0. Exit");
            System.out.print("Select: ");
            choice = sc.nextInt();

            switch (choice)
            {
                case 1:
                    System.out.println("\nStudent List:");
                    for (Student s : students) s.printStudent();
                    break;

                case 2:
                    System.out.println("\nBook List:");
                    for (Book b : books) b.printBook();
                    break;

                case 3:
                    System.out.println("\nLoan Data:");
                    for (BookLending bl : loans) bl.printLending();
                    break;

                case 4:
                    BookLending[] sorted = loans.clone();
                    insertionSort(sorted);
                    System.out.println("\nAfter sorting (Largest fine):");
                    for (BookLending bl : sorted) bl.printLending();
                    break;

                case 5:
                    System.out.print("Enter NIM: ");
                    String nim = sc.next();
                    binarySearchByNIM(loans, nim);
                    break;

                case 0:
                    System.out.println("Goodbye!");
                    break;

                default:
                    System.out.println("Invalid option.");
            }
        }
        while (choice != 0);

        sc.close();
    }
}
```

```
=== JTI LIBRARY BORROWING SYSTEM ===
1. Display Students
2. Display Books
3. Display Borrowings
4. Sort by Fines
5. Search by Student ID
0. Exit
Select: 1

Student List:
NIM: 22001 | Name: Andi | Study program: Informatics Engineering
NIM: 22002 | Name: Budi | Study program: Informatics Engineering
NIM: 22003 | Name: Citra | Study program: Business Information Systems

=== JTI LIBRARY BORROWING SYSTEM ===
1. Display Students
2. Display Books
3. Display Borrowings
4. Sort by Fines
5. Search by Student ID
0. Exit
Select: 2

Book List:
Code: B001 | Title: Algorithm | Year: 2020
Code: B002 | Title: Database | Year: 2019
Code: B003 | Title: Programming | Year: 2021
Code: B004 | Title: Physics | Year: 2024

=== JTI LIBRARY BORROWING SYSTEM ===
1. Display Students
2. Display Books
3. Display Borrowings
4. Sort by Fines
5. Search by Student ID
0. Exit
Select: 3

Loan Data:
Andi | Algorithm | Loan Period (days): 7 | Late return of books: 2 | fine: 4000
Budi | Database | Loan Period (days): 3 | Late return of books: 0 | fine: 0
Citra | Programming | Loan Period (days): 10 | Late return of books: 5 | fine: 10000
Citra | Physics | Loan Period (days): 6 | Late return of books: 1 | fine: 2000
Andi | Database | Loan Period (days): 4 | Late return of books: 0 | fine: 0
```

```
=== JTI LIBRARY BORROWING SYSTEM ===
1. Display Students
2. Display Books
3. Display Borrowings
4. Sort by Fines
5. Search by Student ID
0. Exit
Select: 4

After sorting (Largest fine):
Citra | Programming | Loan Period (days): 10 | Late return of books: 5 | fine: 10000
Andi | Algorithm | Loan Period (days): 7 | Late return of books: 2 | fine: 4000
Citra | Physics | Loan Period (days): 6 | Late return of books: 1 | fine: 2000
Budi | Database | Loan Period (days): 3 | Late return of books: 0 | fine: 0
Andi | Database | Loan Period (days): 4 | Late return of books: 0 | fine: 0

=== JTI LIBRARY BORROWING SYSTEM ===
1. Display Students
2. Display Books
3. Display Borrowings
4. Sort by Fines
5. Search by Student ID
0. Exit
Select: 5
Enter NIM: 22003
Citra | Programming | Loan Period (days): 10 | Late return of books: 5 | fine: 10000
Citra | Physics | Loan Period (days): 6 | Late return of books: 1 | fine: 2000

=== JTI LIBRARY BORROWING SYSTEM ===
1. Display Students
2. Display Books
3. Display Borrowings
4. Sort by Fines
5. Search by Student ID
0. Exit
Select: 0
Goodbye!
```

Github:
https://github.com/guramedadar/Praktikum-Algoritma-Struktur-Data/tree/main/src/com/cm1